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#PROCESAMIENTO SEÑALES LAB 7

# Importamos las librerías que usaremos
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scipy.signal as signal
from google.colab import files #esto sirve para subir los archivos txt

#funcion conversión de señales txt eeg a microvoltios
def bitalino_eeg_to_uv(adc_data, vcc=3.3, bits=10, gain=40000):
    #Fórmula oficial de la hoja de datos del sensor EEG de BITalino
    eeg_uv = (((adc_data / (2**bits)) - 0.5) * vcc / gain) * 1e6
    return eeg_uv

fs = 1000 #Frecuencia de muestreo sugerida en la guía de laboratorio (1000 Hz)
nyq = 0.5 * fs

#-----

#Diseño de filtros

#Filtro Pasabanda Butterworth (0.8 - 48 Hz)
low_cut = 0.8 / nyq
high_cut = 48.0 / nyq
b_band, a_band = signal.butter(4, [low_cut, high_cut], btype='bandpass')

#Filtro Notch para ruido de la red eléctrica (60 Hz)
f0 = 60.0
Q = 30.0
b_notch, a_notch = signal.iirnotch(f0, Q, fs)

#Carga de datos en bucle
print("Sube tus archivos .txt extraídos de OpenSignals: \n")
uploaded = files.upload()

for filename in uploaded.keys():
    print(f"\n Procesando y graficando: {filename} ")

    #Leer el .txt ignorando las líneas de configuración que inician con '#'
    df = pd.read_csv(filename, sep='\t', comment='#', header=None)

    #Extraer columnas (A3 = Fp1 en índice 5) y (A4 = Fp2 en índice 6)

    #Fp1 y Fp2 son las señales capturadas en las 2 cejas, para después compar

    fp1_raw = df.iloc[:, 5].values
    fp2_raw = df.iloc[:, 6].values
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#Convertir a Microvoltios (conversión adc)
fp1_uv = bitalino_eeg_to_uv(fp1_raw)
fp2_uv = bitalino_eeg_to_uv(fp2_raw)

N = len(fp1_uv)
Ts = 1 / fs
t = np.arange(N) * Ts

#Aplicar filtros en cascada (Pasabanda -> Notch) usando filtfilt para no (
fp1_filtrada = signal.filtfilt(b_band, a_band, fp1_uv)
fp1_filtrada = signal.filtfilt(b_notch, a_notch, fp1_filtrada)

fp2_filtrada = signal.filtfilt(b_band, a_band, fp2_uv)
fp2_filtrada = signal.filtfilt(b_notch, a_notch, fp2_filtrada)

# -----

#ploteo
plt.figure(figsize=(16, 8))

#Subplot 1: Fp1 Cruda (Arriba Izquierda)
plt.subplot(2, 2, 1)
plt.plot(t, fp1_uv, color='indianred', alpha=0.8)
plt.title(f"{filename} - Fp1 (Señal Cruda)")
plt.ylabel("Amplitud ( $\mu V$ )")
plt.grid(ls=":")

#Subplot 2: Fp1 Filtrada (Arriba Derecha)
plt.subplot(2, 2, 2)
plt.plot(t, fp1_filtrada, color='teal')
plt.title(f"{filename} - Fp1 (Filtrada 0.8-48Hz + Notch 60Hz)")
plt.ylabel("Amplitud ( $\mu V$ )")
plt.grid(ls=":")

#Subplot 3: Fp2 Cruda (Abajo Izquierda)
plt.subplot(2, 2, 3)
plt.plot(t, fp2_uv, color='chocolate', alpha=0.8)
plt.title(f"{filename} - Fp2 (Señal Cruda)")
plt.xlabel("Tiempo (s)")
plt.ylabel("Amplitud ( $\mu V$ )")
plt.grid(ls=":")

#Subplot 4: Fp2 Filtrada (Abajo Derecha)
plt.subplot(2, 2, 4)
plt.plot(t, fp2_filtrada, color='darkblue')
plt.title(f"{filename} - Fp2 (Filtrada 0.8-48Hz + Notch 60Hz)")
plt.xlabel("Tiempo (s)")
plt.ylabel("Amplitud ( $\mu V$ )")

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plt.grid(ls=":")

plt.tight_layout() # Evita que los textos se superpongan
plt.show()
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Sube tus archivos .txt extraídos de OpenSignals:

```
<>:77: SyntaxWarning: invalid escape sequence '\m'
<>:84: SyntaxWarning: invalid escape sequence '\m'
<>:92: SyntaxWarning: invalid escape sequence '\m'
<>:100: SyntaxWarning: invalid escape sequence '\m'
<>:77: SyntaxWarning: invalid escape sequence '\m'
<>:84: SyntaxWarning: invalid escape sequence '\m'
<>:92: SyntaxWarning: invalid escape sequence '\m'
<>:100: SyntaxWarning: invalid escape sequence '\m'
/tmp/ipykernel_1695/3265543668.py:77: SyntaxWarning: invalid escape sequence
plt.ylabel("Amplitud ($\mu$V)")
/tmp/ipykernel_1695/3265543668.py:84: SyntaxWarning: invalid escape sequence
plt.ylabel("Amplitud ($\mu$V)")
/tmp/ipykernel_1695/3265543668.py:92: SyntaxWarning: invalid escape sequence
plt.ylabel("Amplitud ($\mu$V)")
/tmp/ipykernel_1695/3265543668.py:100: SyntaxWarning: invalid escape sequence
plt.ylabel("Amplitud ($\mu$V)")
```

Examinar... 7 archivos seleccionados.

libreprguntadificil.txt(text/plain) - 895531 bytes, last modified: n/a - 100% done

librepreguntafacil.txt(text/plain) - 680410 bytes, last modified: n/a - 100% done

basal3.txt(text/plain) - 1003794 bytes, last modified: n/a - 100% done

artefactos.txt(text/plain) - 905874 bytes, last modified: n/a - 100% done

basal2.txt(text/plain) - 815998 bytes, last modified: n/a - 100% done

ojosfijopunto.txt(text/plain) - 1665913 bytes, last modified: n/a - 100% done

basal1.txt(text/plain) - 1302570 bytes, last modified: n/a - 100% done

Saving libreprguntadificil.txt to libreprguntadificil (1).txt

Saving librepreguntafacil.txt to librepreguntafacil (2).txt

Saving basal3.txt to basal3 (2).txt

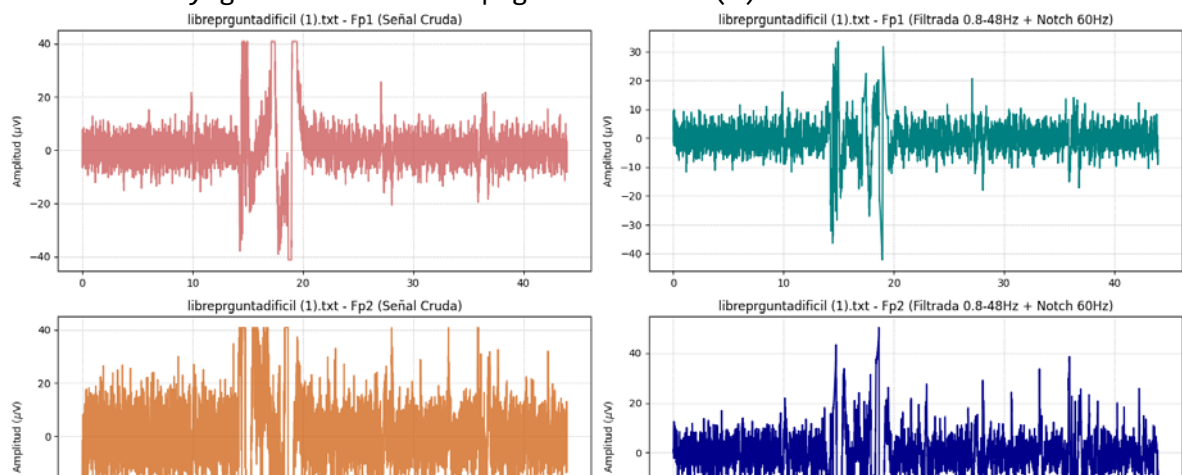
Saving artefactos.txt to artefactos (2).txt

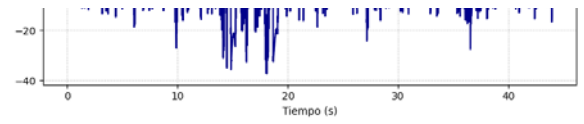
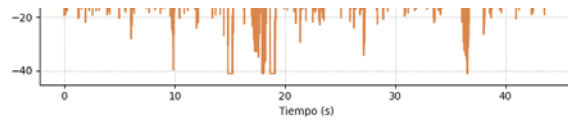
Saving basal2.txt to basal2 (2).txt

Saving ojosfijopunto.txt to ojosfijopunto (2).txt

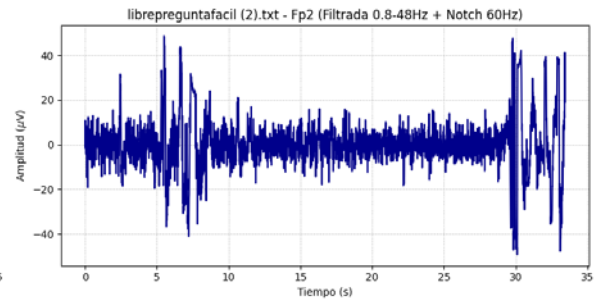
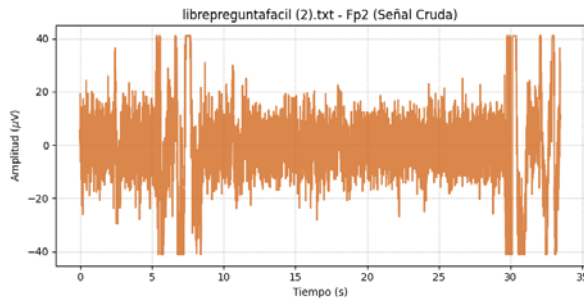
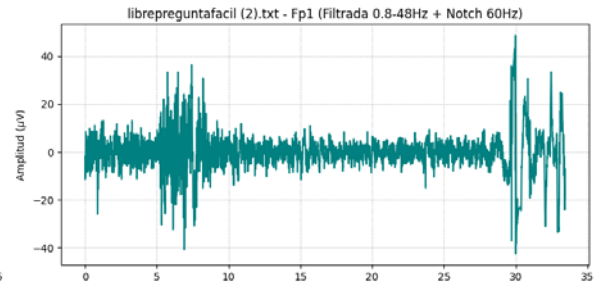
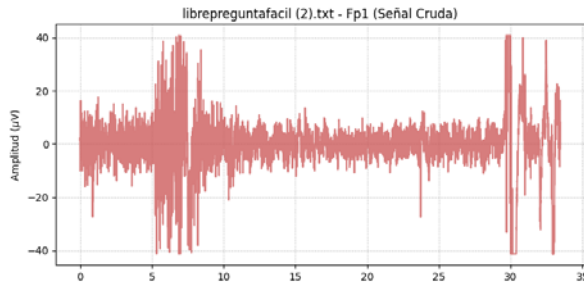
Saving basal1.txt to basal1 (2).txt

Procesando y graficando: libreprguntadificil (1).txt

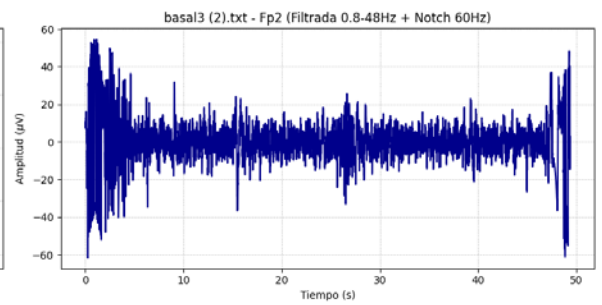
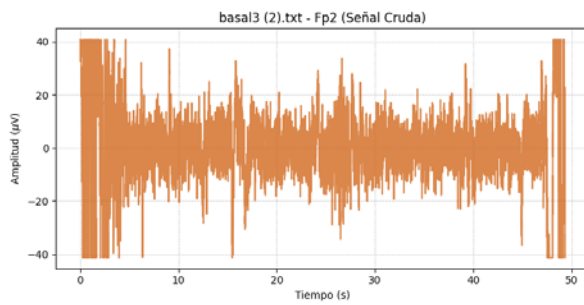
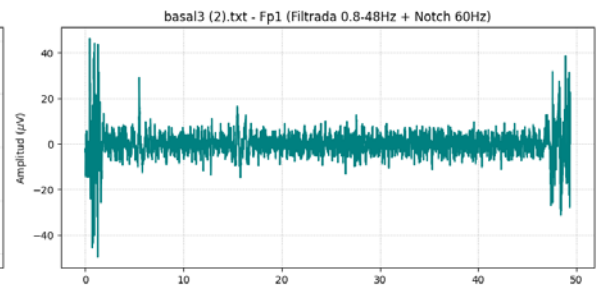
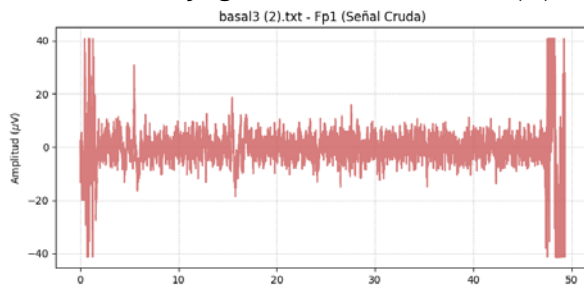




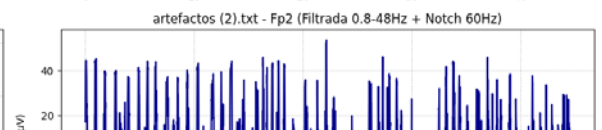
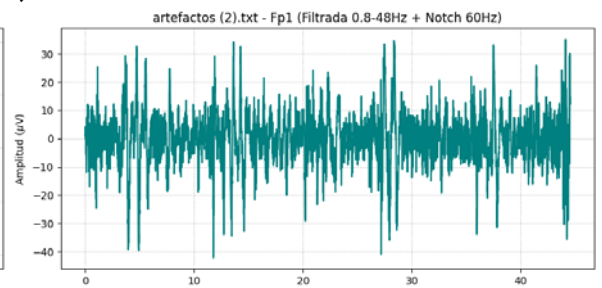
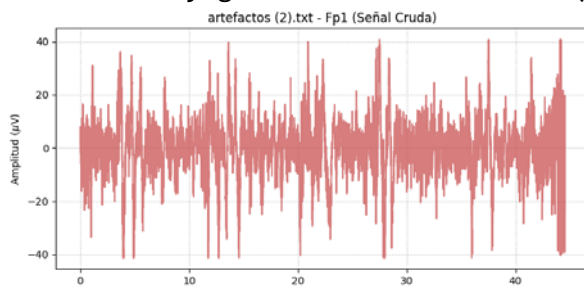
Procesando y graficando: librepreguntafacil (2).txt

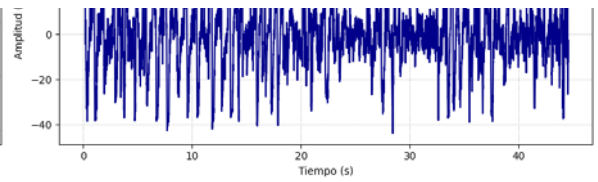
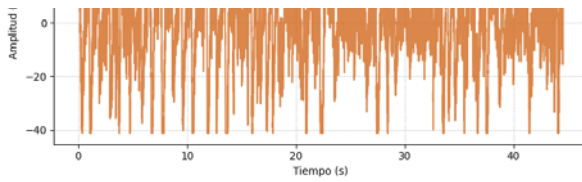


Procesando y graficando: basal3 (2).txt

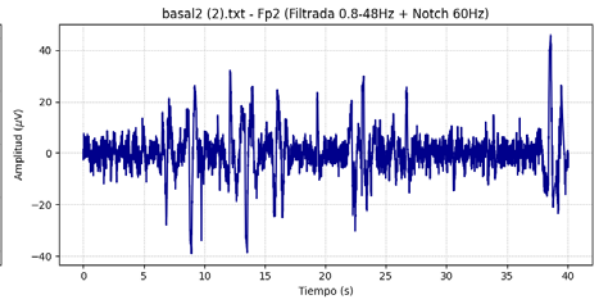
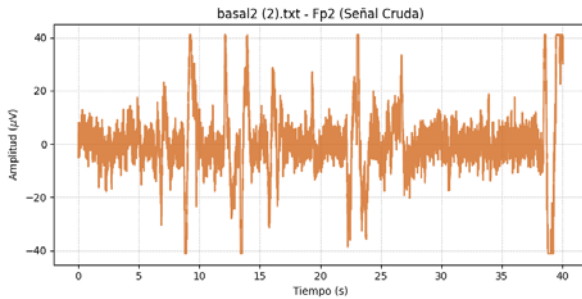
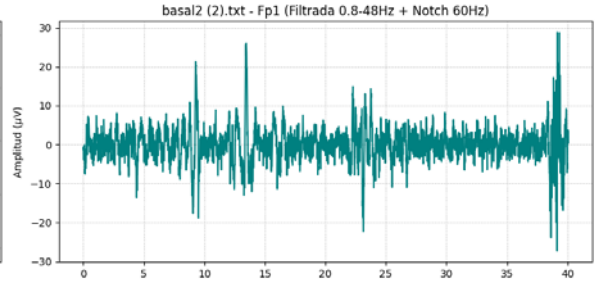
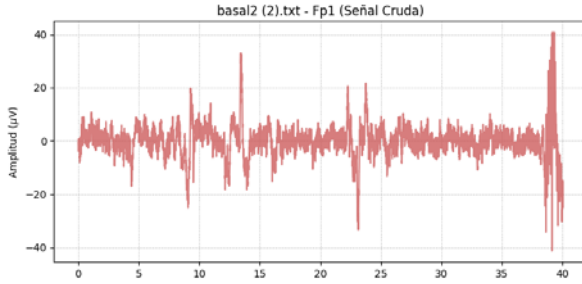


Procesando y graficando: artefactos (2).txt

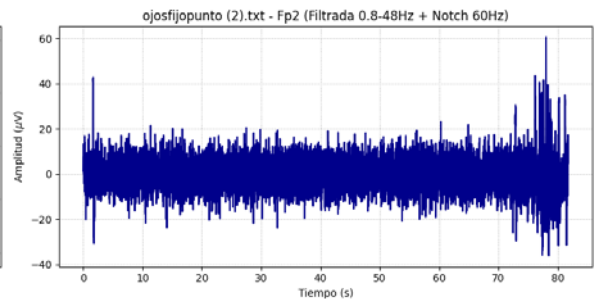
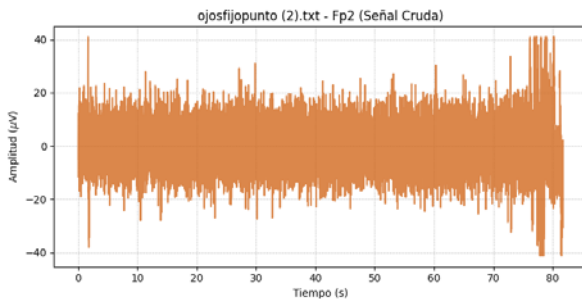
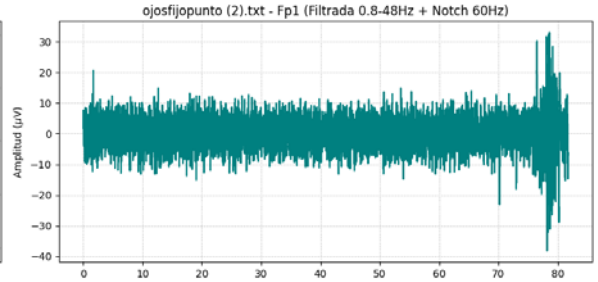
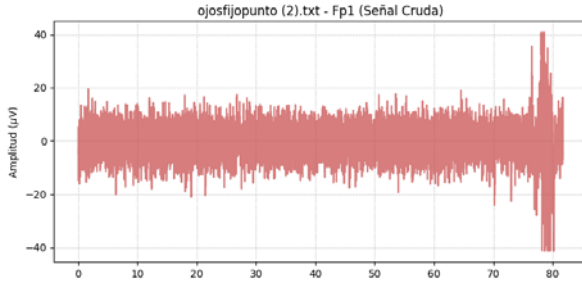




Procesando y graficando: basal2 (2).txt



Procesando y graficando: ojosfijopunto (2).txt



Procesando y graficando: basal1 (2).txt

